Golomb Rulers - 2

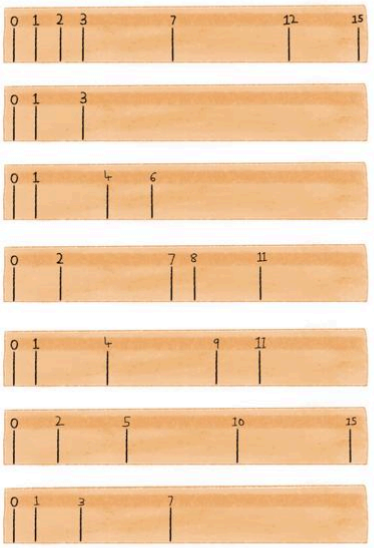
Grade 8: Number Play

Introduction

An interesting function of computers is to make them faster by figuring out ways to perform a task with the minimum amount of information. We need to develop efficient, fast methods to solve problems, but we also need to optimise the amount of information required to complete a task. The less information needed, the less computer memory required, the more efficient the overall process is.

In this activity, we will explore the idea of minimising the amount of information needed to perform a task by looking at an interesting type of ruler known as a Golomb ruler. The speciality of these rulers is that even though they do not have all the markings in a typical ruler, they can still measure all the integer lengths uniquely. The task the rulers perform is measuring integer lengths; a smaller number of markings corresponds to less information needed to do the task.





Activity	Time	Description
Launch	5 min	The teacher shows students the rulers and introduces the idea of Golomb rulers.
Problem-solving and algorithm	25 mins	Student's Tryout the activity with each other. Student Worksheet: https://tinyurl.com/goulombruler1worksheet

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Activity	Time	Description
		As the teacher progresses through the activities, the students progress along with the worksheet.
Discussions and Explorations	5 min	The discussion on the various algorithms, how a slight change in approach leads to a faster solution, and how computers apply such algorithms

CT Connection

Pattern Recognition: Students observe patterns, such as the number of distances formed from a given number of marks and the relationship between order and the number of pairwise distances.

Algorithmic Thinking: Students develop a step-by-step method (algorithm) to test whether a ruler is Golomb by listing all pairs of markings, computing distances, storing distances (mentally or in a list), checking if all distances are unique, etc.

Decomposition: Students break down the problem of verifying a Golomb ruler into smaller, manageable steps, identify all pairs of markings, and calculate distances between each pair of markings

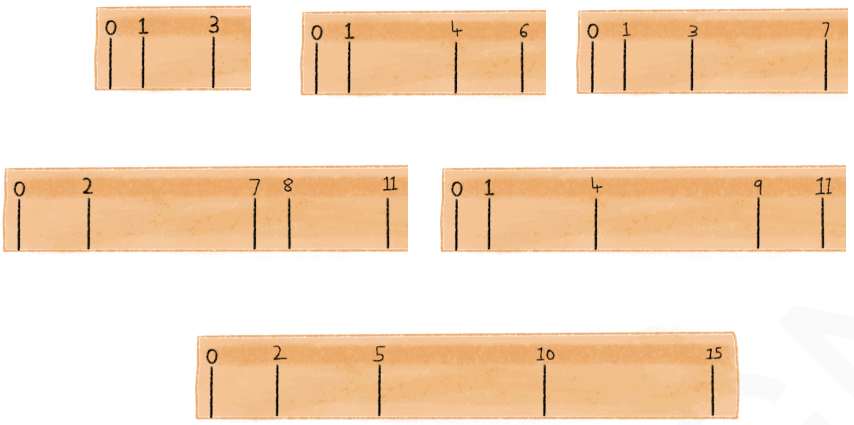
Abstraction: They ignore physical ruler details and think of the problem numerically, enabling general reasoning.

Generalisation: Students extend their understanding to any set of numbers forming a ruler. Find a general formula for the number of distances as the sum of natural numbers.

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Activity

Let us make the following Rulers on strips of paper.



Let us understand the ruler first. Let's say the largest mark on the ruler is 6, and the number of marked lengths on it is 4 (assuming the first mark is at 0). We say the ruler has a length of 6 units and an order of 4. We call this a Golomb ruler if the distances between all pairs of marks are unique. We call a Golomb Ruler a Perfect Golomb Ruler if it measures all the integer lengths up to the length of the ruler.

Is (0, 1, 3) a Golomb Ruler?



Let's say we have a ruler that has a length of 3 and an order of 3. The marked positions are 0, 1, and 3. Can we call the ruler (0, 1, 3) a Golomb ruler?

1. What are all the lengths we can measure using this ruler?
   We can measure lengths
   a. 1 unit
   b. 1 unit and 3 units
   c. 1 unit, 2 units, and 3 units

Answer: c.
Explanation: the distance between the markings 0 to 1 is 1 unit, 1 to 3 is 2 units, and 0 to 3 is 3 units.

2. Is the ruler (0, 1, 3) a Perfect Golomb ruler?

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Answer: Yes
Explanation: The ruler satisfies all the conditions of being a Perfect Golomb ruler; the distances between all the pairs of marks are distinct, and all the possible distances up to the length of the ruler can be measured.

Is the ruler (0, 1, 4, 6) a Golomb ruler?



If we have a ruler with the markings (0, 1, 4, 6), its length is 6 units. Its order is 4.

3. How many distances are there in this ruler?
    a. 4
    b. 5
    c. 6
    d. 7

Answer: c.
Explanation: There are 6 pairs possible when we have to choose a pair of objects from a group of 4 objects: (0,1),(0,4),(0,6),(1,4),(1,6),(4,6).

4. Is this a Perfect Golomb ruler?
    a. Yes, we can measure all the lengths from 1 to 6 units
    b. No, we can't measure all the lengths from 1 to 6 units
    c. Yes, we can measure many integer lengths in a unique way
    d. No, we can't measure some of the integer lengths in unique ways

Answer: a. Yes, we can measure all the lengths from 1 to 6 units
Explanation: This measures all 6 lengths up to the ruler's length in a unique manner.
$1 - 0 = 1\,units$, $6 - 4 = 2\,units$, $4 - 1 = 3\,units$, $4 - 0 = 4\,units$, $6 - 1 = 5\,units$, $6 - 0 = 6\,units$. There are only 6 possible distinct pairs of numbers, and each corresponds to a unique length.

Is the ruler (0, 1, 4, 9, 11) a Golomb ruler?

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The length of the ruler is 10, and the order is 5. Since there are 5 numbers from which we have to choose 2 at a time, there are 10 distinct pairs: (0,1), (0,4), (0,9), (0,11), (1,4), (1,9), (1,11), (4,9), (4,11), (9,11).

5. Is this a Golomb ruler?
    a. Yes, we can measure all the lengths from 1 to 11 units
    b. No, we can't measure all the lengths from 1 to 11 units
    c. Yes, we can measure many integer lengths in a unique way
    d. No, we can't measure some of the integer lengths in unique ways

Answer: c. Yes, we can measure many integer lengths in a unique way
Explanation:
$1 - 0 = 1\,units,\ 4 - 0 = 4\,units, 9 - 0 = 9\,units,\ 11 - 0 = 11\,units$
$4 - 1 = 3\,units, 9 - 1 = 8\,units,\ 11 - 1 = 10\,units$
$9 - 4 = 5\,units,\ 11 - 4 = 7\,units,$ and $11 - 9 = 2\,units$. The distances between all 10 pairs are unique.

6. If this is a Golomb ruler, can this be a Perfect Golomb ruler?
    a. Yes, we can measure all the lengths from 1 to 11 units
    b. No, we can't measure all the lengths from 1 to 11 units
    c. Yes, we can measure many integer lengths in a unique way
    d. No, we can't measure some of the integer lengths in unique ways

Answer: b. No, we can't measure all the lengths from 1 to 11 units
Explanation: The length of the ruler is 11. For this to be a Perfect Golomb ruler, it needs to measure 11 distinct lengths, but the maximum possible distinct distances between points are 10, which is less than the length of the ruler.

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Concept Check

7. If a < b < c < d are markings on a Golomb ruler, then which of the following relations between them is never possible?
    a. $a + d = b + c$
    b. $a + c < b + d$ or vice-versa
    c. $b \times c > a \times d$

Answer: a.
Explanation: $a + d \neq b + c$.
Rather, the sum of a pair cannot be equal to the sum of any other pair. If any such sums are equal, for example, $a + d = c + b$ it implies $d - b = c - a$ which implies distance between the two pairs is equal, a violation of the Golomb property.

8. Which of the given rulers is not a Golomb ruler: (0, 1, 4, 9, 11), (0, 2, 5, 10, 15)?
    a. First
    b. Second
    c. Both
    d. None





Answer: b.
Explanation: When we list all the pairs and their distances for both rulers, we will notice that in the second ruler, the distances repeat. For example
$15 - 10 = 5 = 10 - 5$.

9. For order 6, how many distinct distances should there be?
    a. 10
    b. 12
    c. 20
    d. 15

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Answer: d.
Explanation: By listing down all possible pairs.
Let the markings be $(0,\ a_1,\ a_2,\ a_3,\ a_4,\ a_5)$.

The pairs will be
$(0, a_1),\ (0, a_2),\ (0, a_3),\ (0, a_4),\ (0, a_5),\ (a_1, a_2),\ (a_1, a_3),\ (a_1, a_4),\ (a_1, a_5),$
$(a_2,\ a_3), (a_2,\ a_4), (a_2,\ a_5), (a_3,\ a_4),\ (a_3,\ a_5), (a_4,\ a_5).$

10. What is the total possible number of distinct distances for a Golomb ruler of order n?
    a. n
    b. $\frac{n^2}{2}$
    c. $\frac{n(n-1)}{2}$
    d. $n - 1$

Correct: c.
Explanation: The total possible distinct lengths in golomb ruler of order n can be calculated as follows. The smallest mark can be paired with all the $n - 1$ marks greater than it. The next mark after the smallest mark can be paired with $n - 2$ marks, and similarly, we see that the number of pairs each successive marks for with marks greater than them are $n - 3$, $n - 4$, $n - 5$, so on till 1. This is the sum of the first $n - 1$ natural numbers or also known as the $(n - 1)^{th}$ Triangular number, which is equal to $\frac{n(n-1)}{2}$.

Optimal Golomb rulers

A Golomb ruler, which has the minimum length for a given order (number of markings), is known as the Optimal Golomb ruler.

11. Is (0,1,3,7) an optimal Golomb ruler?



Answer: No
Explanation: The order of the given Golomb ruler is 4 (four markings on the ruler), and its length is 7 units. Remember, we have already seen a Golomb ruler of order 4, (0, 1, 4, 6), whose length is 6 units. We have a Golomb ruler of the same order but lesser length.

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For a given order, there may be more than one optimal ruler. Examples of these optimal rulers for order 5 are (0, 1, 4, 9, 11) and (0, 2, 7, 8, 11). This is possible for Golomb rulers of other orders.

Challenge: Identify some orders for which multiple Optimal Golomb rulers are possible.

We saw that to measure integer length, one doesn't always need a ruler with all markings. We explored the ideas of counting, understanding number sequences, and analyzing whether a mathematical object satisfies the required conditions.

We saw that if we cleverly design things, we can remove much of the information, making our tools more efficient in terms of memory usage. Thinking about how to optimize memory use while solving problems is an important consideration when designing our interactions with computers.

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